PG No. 45 $C_{6h}(c)$ $6/m$ [hexagonal] (internal polar quadrupole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(2,2)}[q](A_g)$

** symmetry

1

** expression

$$-\frac{\sqrt{5}Q_u\left(x^2+y^2-2z^2\right)}{10}+\frac{\sqrt{15}Q_v\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{15}Q_{xy}xy}{5}+\frac{\sqrt{15}Q_{xz}xz}{5}+\frac{\sqrt{15}Q_{yz}yz}{5}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(2,1)}[q](A_g)$

** symmetry

z

** expression

$$\frac{\sqrt{30}Q_vxy}{5}-\frac{\sqrt{30}Q_{xy}\left(x-y\right)\left(x+y\right)}{10}+\frac{\sqrt{30}Q_{xz}yz}{10}-\frac{\sqrt{30}Q_{yz}xz}{10}$$

$\vec{\mathbb{G}}_1^{(2,1)}[q](E_{1g}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$\frac{3\sqrt{5}iQ_uz\left(x+iy\right)}{10}-\frac{\sqrt{15}iQ_vz\left(x-iy\right)}{10}+\frac{\sqrt{15}Q_{xy}z\left(x-iy\right)}{10}+\frac{\sqrt{15}iQ_{xz}\left(x^2+ixy-z^2\right)}{10}+\frac{\sqrt{15}iQ_{yz}\left(xy+iy^2-iz^2\right)}{10}$$

$\vec{\mathbb{G}}_1^{(2,1)}[q](E_{1g}^{(b)})$

** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{5}iQ_uz\left(x-iy\right)}{10}+\frac{\sqrt{15}iQ_vz\left(x+iy\right)}{10}+\frac{\sqrt{15}Q_{xy}z\left(x+iy\right)}{10}-\frac{\sqrt{15}iQ_{xz}\left(x^2-ixy-z^2\right)}{10}-\frac{\sqrt{15}iQ_{yz}\left(xy-iy^2+iz^2\right)}{10}$$

$\vec{\mathbb{Q}}_1^{(2,0)}[q](A_u)$

** symmetry

z

** expression

$$\frac{\sqrt{10}Q_uz}{5}+\frac{\sqrt{30}Q_{xz}x}{10}+\frac{\sqrt{30}Q_{yz}y}{10}$$

$\vec{\mathbb{Q}}_1^{(2,2)}[q](A_u)$

** symmetry

z

** expression

$$-\frac{3\sqrt{35}Q_uz\left(3x^2+3y^2-2z^2\right)}{70}+\frac{\sqrt{105}Q_vz\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}Q_{xy}xyz}{7}-\frac{\sqrt{105}Q_{xz}x\left(x^2+y^2-4z^2\right)}{35}-\frac{\sqrt{105}Q_{yz}y\left(x^2+y^2-4z^2\right)}{35}$$

$\vec{\mathbb{Q}}_1^{(2,0)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{5}Q_u\left(x+iy\right)}{10}+\frac{\sqrt{15}Q_v\left(x-iy\right)}{10}+\frac{\sqrt{15}iQ_{xy}\left(x-iy\right)}{10}+\frac{\sqrt{15}Q_{xz}z}{10}+\frac{\sqrt{15}iQ_{yz}z}{10}$$

$\vec{\mathbb{Q}}_1^{(2,2)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{70}Q_u\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{140}+\frac{\sqrt{210}Q_v\left(3x^3+7ix^2y-7xy^2-2xz^2-3iy^3+2iyz^2\right)}{140}$$
$$-\frac{\sqrt{210}iQ_{xy}\left(x^3+4ix^2y-4xy^2+xz^2-iy^3-iyz^2\right)}{70}+\frac{\sqrt{210}Q_{xz}z\left(4x^2+5ixy-y^2-z^2\right)}{70}-\frac{\sqrt{210}iQ_{yz}z\left(x^2+5ixy-4y^2+z^2\right)}{70}$$

$\vec{\mathbb{Q}}_1^{(2,0)}[q](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{\sqrt{5}Q_u\left(x-iy\right)}{10}+\frac{\sqrt{15}Q_v\left(x+iy\right)}{10}-\frac{\sqrt{15}iQ_{xy}\left(x+iy\right)}{10}+\frac{\sqrt{15}Q_{xz}z}{10}-\frac{\sqrt{15}iQ_{yz}z}{10}$$

$\vec{\mathbb{Q}}_1^{(2,2)}[q](E_{1u}^{(b)})$
** symmetry

$$\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$$

** expression

$$-\frac{3\sqrt{70}Q_u\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{140}+\frac{\sqrt{210}Q_v\left(3x^3-7ix^2y-7xy^2-2xz^2+3iy^3-2iyz^2\right)}{140}$$
$$+\frac{\sqrt{210}iQ_{xy}\left(x^3-4ix^2y-4xy^2+xz^2+iy^3+iyz^2\right)}{70}+\frac{\sqrt{210}Q_{xz}z\left(4x^2-5ixy-y^2-z^2\right)}{70}+\frac{\sqrt{210}iQ_{yz}z\left(x^2-5ixy-4y^2+z^2\right)}{70}$$

* Harmonics for rank 2
$\vec{\mathbb{Q}}_2^{(2,-2)}[q](A_g)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$Q_u$$

$\vec{\mathbb{Q}}_2^{(2,0)}[q](A_g)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{14}Q_u\left(x^2+y^2-2z^2\right)}{14}-\frac{\sqrt{42}Q_v\left(x-y\right)\left(x+y\right)}{14}-\frac{\sqrt{42}Q_{xy}xy}{7}+\frac{\sqrt{42}Q_{xz}xz}{14}+\frac{\sqrt{42}Q_{yz}yz}{14}$$

$\vec{\mathbb{Q}}_2^{(2,2)}[q](A_g)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$\frac{\sqrt{14}Q_u\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{56}-\frac{5\sqrt{42}Q_v\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{168}$$
$$-\frac{5\sqrt{42}Q_{xy}xy\left(x^2+y^2-6z^2\right)}{84}-\frac{5\sqrt{42}Q_{xz}xz\left(3x^2+3y^2-4z^2\right)}{84}-\frac{5\sqrt{42}Q_{yz}yz\left(3x^2+3y^2-4z^2\right)}{84}$$

$\vec{\mathbb{Q}}_2^{(2,-2)}[q](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{2}Q_{xz}}{2}+\frac{\sqrt{2}iQ_{yz}}{2}$$

$\vec{\mathbb{Q}}_2^{(2,0)}[q](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{21}Q_u z\,(x+iy)}{14}+\frac{3\sqrt{7}Q_v z\,(x-iy)}{14}+\frac{3\sqrt{7}iQ_{xy} z\,(x-iy)}{14}+\frac{\sqrt{7}Q_{xz}\left(x^2+3ixy-2y^2+z^2\right)}{14}-\frac{\sqrt{7}iQ_{yz}\left(2x^2+3ixy-y^2-z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,2)}[q](E_{1g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$-\frac{5\sqrt{21}Q_u z\,(x+iy)\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{7}Q_v z\left(5x^3+9ix^2y-9xy^2-2xz^2-5iy^3+2iyz^2\right)}{84}$$
$$-\frac{5\sqrt{7}iQ_{xy} z\left(x^3+6ix^2y-6xy^2+xz^2-iy^3-iyz^2\right)}{42}-\frac{\sqrt{7}Q_{xz}\left(4x^4+5ix^3y+3x^2y^2-27x^2z^2+5ixy^3-30ixyz^2-y^4+3y^2z^2+4z^4\right)}{42}$$
$$+\frac{\sqrt{7}iQ_{yz}\left(x^4+5ix^3y-3x^2y^2-3x^2z^2+5ixy^3-30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$\vec{\mathbb{Q}}_2^{(2,-2)}[q](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{2}Q_{xz}}{2}-\frac{\sqrt{2}iQ_{yz}}{2}$$

$\vec{\mathbb{Q}}_2^{(2,0)}[q](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{21}Q_u z\,(x-iy)}{14}+\frac{3\sqrt{7}Q_v z\,(x+iy)}{14}-\frac{3\sqrt{7}iQ_{xy} z\,(x+iy)}{14}+\frac{\sqrt{7}Q_{xz}\left(x^2-3ixy-2y^2+z^2\right)}{14}+\frac{\sqrt{7}iQ_{yz}\left(2x^2-3ixy-y^2-z^2\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,2)}[q](E_{1g}^{(b)})$
** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$-\frac{5\sqrt{21}Q_u z\,(x-iy)\left(3x^2+3y^2-4z^2\right)}{84}+\frac{5\sqrt{7}Q_v z\left(5x^3-9ix^2y-9xy^2-2xz^2+5iy^3-2iyz^2\right)}{84}$$
$$+\frac{5\sqrt{7}iQ_{xy} z\left(x^3-6ix^2y-6xy^2+xz^2+iy^3+iyz^2\right)}{42}-\frac{\sqrt{7}Q_{xz}\left(4x^4-5ix^3y+3x^2y^2-27x^2z^2-5ixy^3+30ixyz^2-y^4+3y^2z^2+4z^4\right)}{42}$$
$$-\frac{\sqrt{7}iQ_{yz}\left(x^4-5ix^3y-3x^2y^2-3x^2z^2-5ixy^3+30ixyz^2-4y^4+27y^2z^2-4z^4\right)}{42}$$

$\vec{\mathbb{Q}}_2^{(2,-2)}[q](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{2}Q_v}{2}-\frac{\sqrt{2}iQ_{xy}}{2}$$

$\vec{\mathbb{Q}}_2^{(2,0)}[q](E_{2g}^{(a)})$
** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{\sqrt{21}Q_u\,(x-iy)^2}{14}+\frac{\sqrt{7}Q_v\left(x^2+y^2-2z^2\right)}{14}-\frac{\sqrt{7}iQ_{xy}\left(x^2+y^2-2z^2\right)}{14}+\frac{3\sqrt{7}Q_{xz} z\,(x-iy)}{14}-\frac{3\sqrt{7}iQ_{yz} z\,(x-iy)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,2)}[q](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{5\sqrt{21}Q_u\left(x-iy\right)^2\left(x^2+y^2-6z^2\right)}{168} + \frac{\sqrt{7}Q_v\left(19x^4-70ix^3y-102x^2y^2-12x^2z^2+70ixy^3+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$+\frac{\sqrt{7}iQ_{xy}\left(8x^4-35ix^3y-54x^2y^2+6x^2z^2+35ixy^3+8y^4+6y^2z^2-2z^4\right)}{84}$$
$$+\frac{5\sqrt{7}Q_{xz}z\left(x-iy\right)\left(5x^2-7ixy-2y^2-2z^2\right)}{84} + \frac{5\sqrt{7}iQ_{yz}z\left(x-iy\right)\left(2x^2-7ixy-5y^2+2z^2\right)}{84}$$

$\vec{\mathbb{Q}}_2^{(2,-2)}[q](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{2}Q_v}{2} + \frac{\sqrt{2}iQ_{xy}}{2}$$

$\vec{\mathbb{Q}}_2^{(2,0)}[q](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{\sqrt{21}Q_u\left(x+iy\right)^2}{14} + \frac{\sqrt{7}Q_v\left(x^2+y^2-2z^2\right)}{14} + \frac{\sqrt{7}iQ_{xy}\left(x^2+y^2-2z^2\right)}{14} + \frac{3\sqrt{7}Q_{xz}z\left(x+iy\right)}{14} + \frac{3\sqrt{7}iQ_{yz}z\left(x+iy\right)}{14}$$

$\vec{\mathbb{Q}}_2^{(2,2)}[q](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{5\sqrt{21}Q_u\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{168} + \frac{\sqrt{7}Q_v\left(19x^4+70ix^3y-102x^2y^2-12x^2z^2-70ixy^3+19y^4-12y^2z^2+4z^4\right)}{168}$$
$$-\frac{\sqrt{7}iQ_{xy}\left(8x^4+35ix^3y-54x^2y^2+6x^2z^2-35ixy^3+8y^4+6y^2z^2-2z^4\right)}{84}$$
$$+\frac{5\sqrt{7}Q_{xz}z\left(x+iy\right)\left(5x^2+7ixy-2y^2-2z^2\right)}{84} - \frac{5\sqrt{7}iQ_{yz}z\left(x+iy\right)\left(2x^2+7ixy-5y^2+2z^2\right)}{84}$$

$\vec{\mathbb{G}}_2^{(2,-1)}[q](A_u)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{2}Q_{xz}y}{2} - \frac{\sqrt{2}Q_{yz}x}{2}$$

$\vec{\mathbb{G}}_2^{(2,1)}[q](A_u)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{5\sqrt{42}Q_vxyz}{14} - \frac{5\sqrt{42}Q_{xy}z\left(x-y\right)\left(x+y\right)}{28} - \frac{\sqrt{42}Q_{xz}y\left(x^2+y^2-4z^2\right)}{28} + \frac{\sqrt{42}Q_{yz}x\left(x^2+y^2-4z^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(2,-1)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{iQ_u\left(x+iy\right)}{2} + \frac{\sqrt{3}iQ_v\left(x-iy\right)}{6} - \frac{\sqrt{3}Q_{xy}\left(x-iy\right)}{6} - \frac{\sqrt{3}iQ_{xz}z}{6} + \frac{\sqrt{3}Q_{yz}z}{6}$$

$\vec{\mathbb{G}}_2^{(2,1)}[q](E_{1u}^{(a)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$$

** expression

$$-\frac{\sqrt{21}iQ_u\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{7}iQ_v\left(x^3+9ix^2y-9xy^2+6xz^2-iy^3-6iyz^2\right)}{28}$$
$$-\frac{\sqrt{7}Q_{xy}\left(2x^3+3ix^2y-3xy^2-3xz^2-2iy^3+3iyz^2\right)}{14}+\frac{\sqrt{7}iQ_{xz}z\left(3x^2+3y^2-2z^2\right)}{14}-\frac{\sqrt{7}Q_{yz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,-1)}[q](E_{1u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$-\frac{iQ_u\left(x-iy\right)}{2}-\frac{\sqrt{3}iQ_v\left(x+iy\right)}{6}-\frac{\sqrt{3}Q_{xy}\left(x+iy\right)}{6}+\frac{\sqrt{3}iQ_{xz}z}{6}+\frac{\sqrt{3}Q_{yz}z}{6}$$

$\vec{\mathbb{G}}_2^{(2,1)}[q](E_{1u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$$

** expression

$$\frac{\sqrt{21}iQ_u\left(x-iy\right)\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{7}iQ_v\left(x^3-9ix^2y-9xy^2+6xz^2+iy^3+6iyz^2\right)}{28}$$
$$-\frac{\sqrt{7}Q_{xy}\left(2x^3-3ix^2y-3xy^2-3xz^2+2iy^3-3iyz^2\right)}{14}-\frac{\sqrt{7}iQ_{xz}z\left(3x^2+3y^2-2z^2\right)}{14}-\frac{\sqrt{7}Q_{yz}z\left(3x^2+3y^2-2z^2\right)}{14}$$

$\vec{\mathbb{G}}_2^{(2,-1)}[q](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{\sqrt{3}iQ_vz}{3}+\frac{\sqrt{3}Q_{xy}z}{3}-\frac{\sqrt{3}iQ_{xz}\left(x-iy\right)}{6}-\frac{\sqrt{3}Q_{yz}\left(x-iy\right)}{6}$$

$\vec{\mathbb{G}}_2^{(2,1)}[q](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{5\sqrt{21}iQ_uz\left(x-iy\right)^2}{28}+\frac{\sqrt{7}iQ_vz\left(3x^2+3y^2-2z^2\right)}{28}+\frac{\sqrt{7}Q_{xy}z\left(3x^2+3y^2-2z^2\right)}{28}$$
$$-\frac{\sqrt{7}iQ_{xz}\left(x-iy\right)\left(4x^2-5ixy-y^2-6z^2\right)}{28}+\frac{\sqrt{7}Q_{yz}\left(x-iy\right)\left(x^2-5ixy-4y^2+6z^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(2,-1)}[q](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$-\frac{\sqrt{3}iQ_vz}{3}+\frac{\sqrt{3}Q_{xy}z}{3}+\frac{\sqrt{3}iQ_{xz}\left(x+iy\right)}{6}-\frac{\sqrt{3}Q_{yz}\left(x+iy\right)}{6}$$

$\vec{\mathbb{G}}_2^{(2,1)}[q](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$$

** expression

$$\frac{5\sqrt{21}iQ_uz\left(x+iy\right)^2}{28}-\frac{\sqrt{7}iQ_vz\left(3x^2+3y^2-2z^2\right)}{28}+\frac{\sqrt{7}Q_{xy}z\left(3x^2+3y^2-2z^2\right)}{28}$$
$$+\frac{\sqrt{7}iQ_{xz}\left(x+iy\right)\left(4x^2+5ixy-y^2-6z^2\right)}{28}+\frac{\sqrt{7}Q_{yz}\left(x+iy\right)\left(x^2+5ixy-4y^2+6z^2\right)}{28}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(2,-1)}[q](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{30}Q_vxy}{10}+\frac{\sqrt{30}Q_{xy}\left(x-y\right)\left(x+y\right)}{20}+\frac{\sqrt{30}Q_{xz}yz}{5}-\frac{\sqrt{30}Q_{yz}xz}{5}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](A_g)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}Q_vxy\left(x^2+y^2-6z^2\right)}{6}+\frac{\sqrt{15}Q_{xy}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{12}-\frac{\sqrt{15}Q_{xz}yz\left(3x^2+3y^2-4z^2\right)}{12}+\frac{\sqrt{15}Q_{yz}xz\left(3x^2+3y^2-4z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{\sqrt{3}Q_vxz}{2}+\frac{\sqrt{3}Q_{xy}yz}{2}+\frac{\sqrt{3}Q_{xz}\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{3}Q_{yz}xy}{2}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{3}Q_vyz}{2}+\frac{\sqrt{3}Q_{xy}xz}{2}-\frac{\sqrt{3}Q_{xz}xy}{2}-\frac{\sqrt{3}Q_{yz}\left(x-y\right)\left(x+y\right)}{4}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](B_g,1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{7\sqrt{2}Q_uxz\left(x^2-3y^2\right)}{8}-\frac{\sqrt{6}Q_vxz\left(3x^2+3y^2-4z^2\right)}{24}+\frac{\sqrt{6}Q_{xy}yz\left(3x^2+3y^2-4z^2\right)}{24}$$
$$+\frac{\sqrt{6}Q_{xz}\left(5x^4-21x^2y^2-9x^2z^2+2y^4+9y^2z^2\right)}{24}+\frac{\sqrt{6}Q_{yz}xy\left(11x^2-17y^2+18z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](B_g,2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{7\sqrt{2}Q_uyz\left(3x^2-y^2\right)}{8}+\frac{\sqrt{6}Q_vyz\left(3x^2+3y^2-4z^2\right)}{24}+\frac{\sqrt{6}Q_{xy}xz\left(3x^2+3y^2-4z^2\right)}{24}$$
$$-\frac{\sqrt{6}Q_{xz}xy\left(17x^2-11y^2-18z^2\right)}{24}+\frac{\sqrt{6}Q_{yz}\left(2x^4-21x^2y^2+9x^2z^2+5y^4-9y^2z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{30}iQ_uz\left(x+iy\right)}{10}+\frac{3\sqrt{10}iQ_vz\left(x-iy\right)}{20}-\frac{3\sqrt{10}Q_{xy}z\left(x-iy\right)}{20}-\frac{\sqrt{10}iQ_{xz}\left(x^2+6ixy-5y^2+4z^2\right)}{40}-\frac{\sqrt{10}Q_{yz}\left(5x^2+6ixy-y^2-4z^2\right)}{40}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](E_{1g}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{15}iQ_uz\left(x+iy\right)\left(3x^2+3y^2-4z^2\right)}{24}-\frac{\sqrt{5}iQ_vz\left(x^3+27ix^2y-27xy^2+8xz^2-iy^3-8iyz^2\right)}{24}$$
$$-\frac{\sqrt{5}Q_{xy}z\left(13x^3+15ix^2y-15xy^2-8xz^2-13iy^3+8iyz^2\right)}{24}-\frac{\sqrt{5}iQ_{xz}\left(x^4-ix^3y+3x^2y^2-9x^2z^2-ixy^3+6ixyz^2+2y^4-15y^2z^2+4z^4\right)}{24}$$
$$+\frac{\sqrt{5}Q_{yz}\left(2x^4+ix^3y+3x^2y^2-15x^2z^2+ixy^3-6ixyz^2+y^4-9y^2z^2+4z^4\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{\sqrt{30}iQ_uz\left(x-iy\right)}{10}-\frac{3\sqrt{10}iQ_vz\left(x+iy\right)}{20}-\frac{3\sqrt{10}Q_{xy}z\left(x+iy\right)}{20}+\frac{\sqrt{10}iQ_{xz}\left(x^2-6ixy-5y^2+4z^2\right)}{40}-\frac{\sqrt{10}Q_{yz}\left(5x^2-6ixy-y^2-4z^2\right)}{40}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](E_{1g}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{15}iQ_uz\left(x-iy\right)\left(3x^2+3y^2-4z^2\right)}{24}+\frac{\sqrt{5}iQ_vz\left(x^3-27ix^2y-27xy^2+8xz^2+iy^3+8iyz^2\right)}{24}$$
$$-\frac{\sqrt{5}Q_{xy}z\left(13x^3-15ix^2y-15xy^2-8xz^2+13iy^3-8iyz^2\right)}{24}+\frac{\sqrt{5}iQ_{xz}\left(x^4+ix^3y+3x^2y^2-9x^2z^2+ixy^3-6ixyz^2+2y^4-15y^2z^2+4z^4\right)}{24}$$
$$+\frac{\sqrt{5}Q_{yz}\left(2x^4-ix^3y+3x^2y^2-15x^2z^2-ixy^3+6ixyz^2+y^4-9y^2z^2+4z^4\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{\sqrt{3}iQ_u\left(x-iy\right)^2}{4}-\frac{iQ_v\left(x^2+y^2-2z^2\right)}{4}-\frac{Q_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](E_{2g}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{6}iQ_u\left(x-iy\right)^2\left(x^2+y^2-6z^2\right)}{12}+\frac{\sqrt{2}iQ_v\left(x^4-7ix^3y-12x^2y^2+6x^2z^2+7ixy^3+y^4+6y^2z^2-2z^4\right)}{12}$$
$$-\frac{\sqrt{2}Q_{xy}\left(5x^4-14ix^3y-18x^2y^2-12x^2z^2+14ixy^3+5y^4-12y^2z^2+4z^4\right)}{24}$$
$$-\frac{7\sqrt{2}iQ_{xz}z\left(x-iy\right)\left(2x^2-ixy+y^2-2z^2\right)}{24}-\frac{7\sqrt{2}Q_{yz}z\left(x-iy\right)\left(x^2+ixy+2y^2-2z^2\right)}{24}$$

$\vec{\mathbb{G}}_3^{(2,-1)}[q](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{3}iQ_u\left(x+iy\right)^2}{4}+\frac{iQ_v\left(x^2+y^2-2z^2\right)}{4}-\frac{Q_{xy}\left(x^2+y^2-2z^2\right)}{4}$$

$\vec{\mathbb{G}}_3^{(2,1)}[q](E_{2g}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{\sqrt{6}iQ_u\left(x+iy\right)^2\left(x^2+y^2-6z^2\right)}{12}-\frac{\sqrt{2}iQ_v\left(x^4+7ix^3y-12x^2y^2+6x^2z^2-7ixy^3+y^4+6y^2z^2-2z^4\right)}{12}$$
$$-\frac{\sqrt{2}Q_{xy}\left(5x^4+14ix^3y-18x^2y^2-12x^2z^2-14ixy^3+5y^4-12y^2z^2+4z^4\right)}{24}$$
$$+\frac{7\sqrt{2}iQ_{xz}z\left(x+iy\right)\left(2x^2+ixy+y^2-2z^2\right)}{24}-\frac{7\sqrt{2}Q_{yz}z\left(x+iy\right)\left(x^2-ixy+2y^2-2z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](A_u)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{15}Q_uz}{5}-\frac{\sqrt{5}Q_{xz}x}{5}-\frac{\sqrt{5}Q_{yz}y}{5}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](A_u)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{15}Q_uz\left(3x^2+3y^2-2z^2\right)}{15}-\frac{\sqrt{5}Q_vz\left(x-y\right)\left(x+y\right)}{2}-\sqrt{5}Q_{xy}xyz-\frac{\sqrt{5}Q_{xz}x\left(x^2+y^2-4z^2\right)}{20}-\frac{\sqrt{5}Q_{yz}y\left(x^2+y^2-4z^2\right)}{20}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](A_u)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{\sqrt{330}Q_uz\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{264}-\frac{7\sqrt{110}Q_vz\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}-\frac{7\sqrt{110}Q_{xy}xyz\left(x^2+y^2-2z^2\right)}{44}$$
$$+\frac{\sqrt{110}Q_{xz}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}+\frac{\sqrt{110}Q_{yz}y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{44}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](B_u,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$\frac{\sqrt{2}Q_vy}{2}+\frac{\sqrt{2}Q_{xy}x}{2}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](B_u,2)$
** symmetry

$$\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$$

** expression

$$\frac{\sqrt{2}Q_vx}{2}-\frac{\sqrt{2}Q_{xy}y}{2}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](B_u,1)$
** symmetry

$$\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{5\sqrt{6}Q_uy\left(3x^2-y^2\right)}{24}+\frac{\sqrt{2}Q_vy\left(x^2+y^2-4z^2\right)}{8}+\frac{\sqrt{2}Q_{xy}x\left(x^2+y^2-4z^2\right)}{8}+\frac{5\sqrt{2}Q_{xz}xyz}{4}+\frac{5\sqrt{2}Q_{yz}z\left(x-y\right)\left(x+y\right)}{8}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](B_u, 2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{5\sqrt{6}Q_u x\left(x^2-3y^2\right)}{24} + \frac{\sqrt{2}Q_v x\left(x^2+y^2-4z^2\right)}{8} - \frac{\sqrt{2}Q_{xy} y\left(x^2+y^2-4z^2\right)}{8} + \frac{5\sqrt{2}Q_{xz} z\left(x-y\right)\left(x+y\right)}{8} - \frac{5\sqrt{2}Q_{yz} xyz}{4}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](B_u, 1)$

** symmetry

$$\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$$

** expression

$$-\frac{7\sqrt{33}Q_u y\left(3x^2-y^2\right)\left(x^2+y^2-8z^2\right)}{264} + \frac{\sqrt{11}Q_v y\left(53x^4-104x^2y^2-6x^2z^2+11y^4-6y^2z^2+4z^4\right)}{88}$$
$$-\frac{\sqrt{11}Q_{xy} x\left(5x^4-53x^2y^2+3x^2z^2+26y^4+3y^2z^2-2z^4\right)}{44} + \frac{7\sqrt{11}Q_{xz} xyz\left(7x^2-5y^2-2z^2\right)}{44} - \frac{7\sqrt{11}Q_{yz} z\left(x^4-9x^2y^2+x^2z^2+2y^4-y^2z^2\right)}{44}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](B_u, 2)$

** symmetry

$$\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$$

** expression

$$-\frac{7\sqrt{33}Q_u x\left(x^2-3y^2\right)\left(x^2+y^2-8z^2\right)}{264} + \frac{\sqrt{11}Q_v x\left(11x^4-104x^2y^2-6x^2z^2+53y^4-6y^2z^2+4z^4\right)}{88}$$
$$+\frac{\sqrt{11}Q_{xy} y\left(26x^4-53x^2y^2+3x^2z^2+5y^4+3y^2z^2-2z^4\right)}{44} + \frac{7\sqrt{11}Q_{xz} z\left(2x^4-9x^2y^2-x^2z^2+y^4+y^2z^2\right)}{44} + \frac{7\sqrt{11}Q_{yz} xyz\left(5x^2-7y^2+2z^2\right)}{44}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](E_{1u}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{5}Q_u\left(x+iy\right)}{5} - \frac{\sqrt{15}Q_v\left(x-iy\right)}{30} - \frac{\sqrt{15}iQ_{xy}\left(x-iy\right)}{30} + \frac{2\sqrt{15}Q_{xz}z}{15} + \frac{2\sqrt{15}iQ_{yz}z}{15}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](E_{1u}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$-\frac{3\sqrt{5}Q_u\left(x+iy\right)\left(x^2+y^2-4z^2\right)}{40} - \frac{\sqrt{15}Q_v\left(11x^3+9ix^2y-9xy^2-24xz^2-11iy^3+24iyz^2\right)}{120}$$
$$-\frac{\sqrt{15}iQ_{xy}\left(x^3-21ix^2y+21xy^2-24xz^2-iy^3+24iyz^2\right)}{120} + \frac{\sqrt{15}Q_{xz}z\left(9x^2+30ixy-21y^2+4z^2\right)}{120} - \frac{\sqrt{15}iQ_{yz}z\left(21x^2+30ixy-9y^2-4z^2\right)}{120}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](E_{1u}^{(a)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$$

** expression

$$\frac{3\sqrt{110}Q_u\left(x+iy\right)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}$$
$$-\frac{\sqrt{330}Q_v\left(5x^5+9ix^4y-4x^3y^2-46x^3z^2+4ix^2y^3-66ix^2yz^2-9xy^4+66xy^2z^2+12xz^4-5iy^5+46iy^3z^2-12iyz^4\right)}{528}$$
$$+\frac{\sqrt{330}iQ_{xy}\left(x^5+6ix^4y-5x^3y^2-5x^3z^2+5ix^2y^3-51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4-iy^5+5iy^3z^2+6iyz^4\right)}{264}$$
$$-\frac{\sqrt{330}Q_{xz}z\left(18x^4+21ix^3y+15x^2y^2-41x^2z^2+21ixy^3-42ixyz^2-3y^4+y^2z^2+4z^4\right)}{264}$$
$$+\frac{\sqrt{330}iQ_{yz}z\left(3x^4+21ix^3y-15x^2y^2-x^2z^2+21ixy^3-42ixyz^2-18y^4+41y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](E_{1u}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\frac{\sqrt{5}Q_u\,(x-iy)}{5}-\frac{\sqrt{15}Q_v\,(x+iy)}{30}+\frac{\sqrt{15}iQ_{xy}\,(x+iy)}{30}+\frac{2\sqrt{15}Q_{xz}z}{15}-\frac{2\sqrt{15}iQ_{yz}z}{15}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](E_{1u}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$-\frac{3\sqrt{5}Q_u\,(x-iy)\left(x^2+y^2-4z^2\right)}{40}-\frac{\sqrt{15}Q_v\left(11x^3-9ix^2y-9xy^2-24xz^2+11iy^3-24iyz^2\right)}{120}$$
$$+\frac{\sqrt{15}iQ_{xy}\left(x^3+21ix^2y+21xy^2-24xz^2+iy^3-24iyz^2\right)}{120}+\frac{\sqrt{15}Q_{xz}z\left(9x^2-30ixy-21y^2+4z^2\right)}{120}+\frac{\sqrt{15}iQ_{yz}z\left(21x^2-30ixy-9y^2-4z^2\right)}{120}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](E_{1u}^{(b)})$

** symmetry

$$-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$$

** expression

$$\frac{3\sqrt{110}Q_u\,(x-iy)\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{176}$$
$$-\frac{\sqrt{330}Q_v\left(5x^5-9ix^4y-4x^3y^2-46x^3z^2-4ix^2y^3+66ix^2yz^2-9xy^4+66xy^2z^2+12xz^4+5iy^5-46iy^3z^2+12iyz^4\right)}{528}$$
$$-\frac{\sqrt{330}iQ_{xy}\left(x^5-6ix^4y-5x^3y^2-5x^3z^2-5ix^2y^3+51ix^2yz^2-6xy^4+51xy^2z^2-6xz^4+iy^5-5iy^3z^2-6iyz^4\right)}{264}$$
$$-\frac{\sqrt{330}Q_{xz}z\left(18x^4-21ix^3y+15x^2y^2-41x^2z^2-21ixy^3+42ixyz^2-3y^4+y^2z^2+4z^4\right)}{264}$$
$$-\frac{\sqrt{330}iQ_{yz}z\left(3x^4-21ix^3y-15x^2y^2-x^2z^2-21ixy^3+42ixyz^2-18y^4+41y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{6}Q_vz}{6}-\frac{\sqrt{6}iQ_{xy}z}{6}+\frac{\sqrt{6}Q_{xz}\,(x-iy)}{6}-\frac{\sqrt{6}iQ_{yz}\,(x-iy)}{6}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{6}Q_vz\left(3x^2+3y^2-2z^2\right)}{12}-\frac{\sqrt{6}iQ_{xy}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{6}Q_{xz}\,(x-iy)\left(x^2-5ixy-4y^2+6z^2\right)}{24}+\frac{\sqrt{6}iQ_{yz}\,(x-iy)\left(4x^2-5ixy-y^2-6z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](E_{2u}^{(a)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$-\frac{21\sqrt{11}Q_uz\,(x-iy)^2\left(x^2+y^2-2z^2\right)}{88}+\frac{\sqrt{33}Q_vz\left(39x^4-126ix^3y-174x^2y^2-20x^2z^2+126ixy^3+39y^4-20y^2z^2+4z^4\right)}{264}$$
$$+\frac{\sqrt{33}iQ_{xy}z\left(12x^4-63ix^3y-102x^2y^2+10x^2z^2+63ixy^3+12y^4+10y^2z^2-2z^4\right)}{132}$$
$$-\frac{\sqrt{33}Q_{xz}\,(x-iy)\left(5x^4-7ix^3y+3x^2y^2-46x^2z^2-7ixy^3+56ixyz^2-2y^4+10y^2z^2+12z^4\right)}{132}$$
$$-\frac{\sqrt{33}iQ_{yz}\,(x-iy)\left(2x^4-7ix^3y-3x^2y^2-10x^2z^2-7ixy^3+56ixyz^2-5y^4+46y^2z^2-12z^4\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(2,-2)}[q](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{6}Q_vz}{6}+\frac{\sqrt{6}iQ_{xy}z}{6}+\frac{\sqrt{6}Q_{xz}\left(x+iy\right)}{6}+\frac{\sqrt{6}iQ_{yz}\left(x+iy\right)}{6}$$

$\vec{\mathbb{Q}}_3^{(2,0)}[q](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\frac{\sqrt{6}Q_vz\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{6}iQ_{xy}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{6}Q_{xz}\left(x+iy\right)\left(x^2+5ixy-4y^2+6z^2\right)}{24}-\frac{\sqrt{6}iQ_{yz}\left(x+iy\right)\left(4x^2+5ixy-y^2-6z^2\right)}{24}$$

$\vec{\mathbb{Q}}_3^{(2,2)}[q](E_{2u}^{(b)})$

** symmetry

$$\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$$

** expression

$$\begin{aligned}&-\frac{21\sqrt{11}Q_uz\left(x+iy\right)^2\left(x^2+y^2-2z^2\right)}{88}+\frac{\sqrt{33}Q_vz\left(39x^4+126ix^3y-174x^2y^2-20x^2z^2-126ixy^3+39y^4-20y^2z^2+4z^4\right)}{264}\\&-\frac{\sqrt{33}iQ_{xy}z\left(12x^4+63ix^3y-102x^2y^2+10x^2z^2-63ixy^3+12y^4+10y^2z^2-2z^4\right)}{132}\\&-\frac{\sqrt{33}Q_{xz}\left(x+iy\right)\left(5x^4+7ix^3y+3x^2y^2-46x^2z^2+7ixy^3-56ixyz^2-2y^4+10y^2z^2+12z^4\right)}{132}\\&+\frac{\sqrt{33}iQ_{yz}\left(x+iy\right)\left(2x^4+7ix^3y-3x^2y^2-10x^2z^2+7ixy^3-56ixyz^2-5y^4+46y^2z^2-12z^4\right)}{132}\end{aligned}$$